

GRAVITYSORT · REQUIREMENTS

GravitySort requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/03-requirements.md @ b7ceb80		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

GravitySort requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs. They will be checked by calculation at TRL 3 and revised after co-design sessions. Recovery can only be verified by testing with real or spiked ore, which is TRL 4 work.

Table 1. Requirements. Status compares the concept estimates in GVS-PRC-001 with each target.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Use no mercury	No amalgamation step, no mercury-coated plates or traps, no step that needs mercury to recover gold from the concentrate	Design review of the flowsheet	Met by design
R2	Process a small group's daily ore	200 kg/h or more of ore milled to below 2 mm, as slurry of 25 to 35 % solids, for 8 h (1.6 t per day)	Bowl capacity and cycle calculation	Met on paper (estimate)
R3	Hold fine gold in the bowl	Bowl acceleration adjustable from 40 to 80 G at the riffle rings (about 600 to 850 rpm at 100 mm radius), with speed shown to the operator	Drive ratio and speed calculation	Met on paper
R4	Recover fine gold	Of free gold in the screened feed: 80 % or more of 75 to 1,000 μm gold and 50 % or more of 38 to 75 μm gold in the bowl concentrate; 60 % or more of all gold in the ore to smelted gold overall	Test with spiked or characterized ore (TRL 4)	Not verifiable at TRL 2 or 3; at risk
R5	Make a concentrate that can be smelted without mercury	Bowl mass pull 0.5 % or less of feed; shaking table reduces one day's bowl concentrate (about 5 kg) to 100 g or less	Mass balance calculation; later test	Met on paper (estimate)
R6	Run on pedal power	Full throughput at 60 W or less at the pedals, at a cadence of 55 to 70 rpm	Power estimate from slurry, fluidization water and bearing losses	Met on paper (about 45 W estimated); at risk until losses are measured
R7	Run on a small motor	Full throughput from the MotionCore 250 W reference drive on a 24 to 48 V pack, with at least 3 times power margin	Power estimate	Met on paper
R8	Use little water	1.5 m ³ /h or less at 200 kg/h, and tolerate recirculated water with fine silt	Water balance calculation	Met on paper (about 1.2 m ³ /h estimated)
R9	Stay within the concept budget	Parts \$350 or less, excluding the MotionCore module and battery	Priced BOM	Not met (about \$437 estimated)
R10	Be built in a local workshop	Welding, drilling and hand tools only; no lathe; bowl liner cast in a printed mold	Design review of every part	Met on paper; bowl casting route unproven
R11	Travel to site	Breaks into loads of 30 kg or less, carried by two people; total 80 kg or less; assembled with hand tools in 30 min or less	Mass estimate from the model	Met on paper (about 75 kg in four loads, estimate)
R12	Guard every moving part	Bowl covered by a lid guard during running; belts, chains and the table head fully guarded; bowl speed limited to 900 rpm or less by gearing (pedal) or by the MotionCore speed limit (motor); bowl stops within 15 s of stopping the drive	Design review and safety checklist	Partly met; bowl brake not yet designed
R13	Clean up quickly and securely	Bowl concentrate flushed into a lockable container in 5 min or less without tools; table concentrate drops into a lockable tray	Design review; later timed trial	Met on paper

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R14	Last in abrasive service	Bowl liner and riffles replaceable in 30 min; liner life 500 h or more (estimate to be checked)	Wear data for cast polyurethane; later test	Not verifiable at TRL 2

■ Assumptions

- Ore is milled to below 2 mm by the user's existing mill and screened on the GravitySort hopper screen.
- Free, liberated gold in the 38 to 1,000 µm range. Gold locked in sulfides or finer than about 20 µm is lost to tailings by any gravity method, so R4 applies to free gold only.
- Reference ore for the daily balance: 5 g/t, 1.6 t per day. Real grades vary widely (about 1 to 20 g/t).
- Healthy adult pedalling at 60 W or less for sessions of about 1 h, with operators taking turns.
- Budget: the \$350 in `project.yaml` covers GravitySort parts. The MotionCore module (about \$325 for its reference kit, per the MotionCore README) and a battery are shared lab components, budgeted with MotionCore. Whether GravitySort's budget should change is proposed, awaiting Amish (see `docs/REVIEW.md`).